PyGameMaker — Tutorial 3: Classic Pong

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Classic Pong: Two-Player Game** (4 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher. You will need a partner to play at the end!

What You Will Make

The classic arcade game: two paddles, a bouncing ball, invisible goals behind each paddle, and a score on the screen. Player 1 uses **W** and **S**; Player 2 uses the **Up** and **Down** arrows.

Phase 1: Paddles and Ball

- ☐ **1.** Create three sprites: spr_ball (16×16, white circle), spr_paddle (16×64, tall rectangle) and spr_wall (32×32, grey block).
- ☐ **2.** Create the objects: obj_wall (sprite spr_wall, **Solid**), obj_paddle_left and obj_paddle_right (sprite spr_paddle, **Solid**), and obj_ball (sprite spr_ball, **not** solid).
- ☐ **3.** Left paddle: **Keyboard: W (held)** with *Set vertical speed to -8*, **Keyboard: S (held)** with *Set vertical speed to 8*, **Keyboard: No key** with *Stop movement*, and **Collision with obj_wall** with *Stop movement*.
- ☐ **4.** Right paddle: the same four events, but with **Up Arrow** and **Down Arrow**.
- ☐ **5.** Ball: **Create** with *Start moving in direction 45 at speed 6*; and three collision events (with obj_wall, obj_paddle_left, obj_paddle_right), each with *Bounce against solid objects*.
- ☐ **6.** Create room_pong (640×480). Place walls along the top and bottom, a paddle near each side edge, and the ball in the centre.

> **You should see:** press **F5**. Both paddles move with their own keys and stop at the walls. The ball flies up and to the right, bounces off walls and paddles, and if you miss it flies off the side of the screen. That is expected for now.

Phase 2: Goals and Scoring

- ☐ **7.** Create a sprite spr_goal (32×32, any colour).
- ☐ **8.** Create obj_goal_left and obj_goal_right (sprite spr_goal, **Visible unchecked**, **Solid** checked).
- ☐ **9.** In obj_ball, add **Collision with obj_goal_left**: *Set variable* global.p2score *relative* +1, then *Jump to start position*. Add **Collision with obj_goal_right**: the same with global.p1score.
- ☐ **10.** In the room, stack obj_goal_left along the left edge and obj_goal_right along the right edge, behind the paddles.

> **You should see:** press **F5**. When the ball passes a paddle it goes back to the centre. You cannot see the score yet.

Phase 3: Score Display

- [ ] **11.** Create obj_score (no sprite). Its **Create** event sets global.p1score and global.p2score to 0.
- [ ] **12.** Its **Draw** event: first *Set draw color* to white (#ffffff); then *Draw text* "Player 1:" at x 10, y 40; *Draw variable* global.p1score at x 100, y 40; *Draw text* "Player 2:" at x 10, y 60; *Draw variable* global.p2score at x 100, y 60.
- [ ] **13.** Place obj_score anywhere in the room.

> **You should see:** press **F5** and play a point with your partner. The scores show at the top left and go up when the other player misses.

Stuck?

What is wrong	Most likely cause	What to do
A paddle goes through the wall	The paddle has no **Collision with obj_wall** event, or obj_wall is not Solid	Add the event (with *Stop movement*); tick **Solid** on obj_wall
The ball goes right through walls or paddles	The Bounce events are missing, or the walls and paddles are not Solid	Add **Bounce against solid objects** for each collision; tick **Solid**
The ball only moves sideways	The direction is not what you expect	Check *Start moving in direction 45* in the ball's **Create** event
The ball is stuck in the middle	The ball's **Create** event is missing, or the ball is not in the room	Add the event; place the ball in the room
The ball never scores	The goals are not in the room, or are not Solid	Place goals behind the paddles; tick **Solid**
The score does not show	obj_score is not in the room, there is no **Draw** event, or the text is black on the black room	Place it; add the Draw event; add *Set draw color* white first
The score is always 0	The variables are called something different in different places	Use exactly global.p1score and global.p2score everywhere
Both players' keys move the same paddle	Both paddles have the same keys	Left: W and S. Right: Up and Down

Challenges

- **Try this (5 minutes):** change the ball speed, or the paddle speed.
- **Push further:** add a dotted line down the middle, or make the ball go faster every time it hits a paddle.
- **Invent:** end the game when a player reaches 10 points and show who won, or change the ball's angle depending on where it hits the paddle.

Vocabulary

Term	What it means
Solid	An object that other objects can bounce off or be stopped by
Bounce	Reverse the direction of movement after hitting something
Global variable	A value shared by all objects, written global.name
Draw event	The event that draws things on the screen every frame
Goal	An (invisible) object that detects when the ball has passed

Check Yourself

Write your answers in the notes below.

1. Why do the goals have **Visible** unchecked but **Solid** checked?
2. What does the global. in global.p1score mean, and why do we need it here?
3. After a goal, the ball is put back in the middle. Which way does it move, and why?

My Notes